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Macroeconomics

Assignment: Chapter 5

1. In an economy, the working-age population is 200 million. Of this total,
- 150.0 million workers are employed.
 - 8.0 million workers are unemployed.
 - 32.0 million workers are not available for work (homemakers, full-time students, etc.).
 - 6.0 million workers are available for work but are discouraged and thus are not seeking work.
 - 4.0 million workers are available for work but are not currently seeking work due to transportation or childcare problems.

The labor force participation rate in this economy is %. (Round your answer to one decimal place.)

2. Why is the unemployment rate, as measured by Statistics Canada, an imperfect measure of the extent of joblessness in the economy?

- ☐ A. Underemployed people are considered employed.
- ☐ B. It fails to account for illegal activities.
- ☐ C. It does not account for inaccurate responses to the Labour Force Survey.
- ☐ D. Discouraged workers are not considered unemployed.
- ☒ E. All of the above.

3. In an economy, the working-age population is 300 million. Of this total,

- 225 million workers are employed.
- 12 million workers are unemployed.
- 48 million workers are not available for work (homemakers, full-time students, etc.).
- 9 million workers are available for work but are discouraged and thus are not seeking work.
- 6 million workers are available for work but are not currently seeking work due to transportation or childcare problems.

The unemployment rate in this economy is %. (Round your answer to one decimal place.)

4. Which of the following is an example of cyclical unemployment?

- ☐ A. Autoworkers lose jobs due to globalization and must retrain to find new jobs.
- ☒ B. During a recession, aircraft assemblers are laid off but expect to be recalled as the economy improves.
- ☐ C. Workers at ski resorts in Colorado are laid off during the summer.
- ☐ D. A worker moves to California and searches for a new job.

5. Which of the following is an example of structural unemployment?

- ☐ A. A worker moves to California and searches for a new job.
- ☐ B. Workers in the tourism industry in Maine are laid off during the winter.
- ☒ C. Autoworkers lose jobs due to globalization and must retrain to find new jobs.
- ☐ D. During a recession, aircraft assemblers are laid off but expect to be recalled as the economy improves.

6. The natural rate of unemployment tends to be lower when

- ☐ A. the minimum wage is higher than the market wage.
- ☒ B. unemployment insurance benefits are lower.
- ☐ C. the efficiency wage is higher.
- ☐ D. labor unions are more effective.

7. The minimum wage in Newfoundland and Labrador was set at only \$0.50 per hour in 1965. This wage was

- ☐ A. deliberately set so low because the government did not want the discouraged workers to join the labor force.
- ☐ B. well below the equilibrium wage since minimum wages are binding price ceilings and are always set below the equilibrium wage rate
- ☒ C. higher than the equilibrium wage since minimum wages are binding price floors and are usually set above the equilibrium wage rate
- ☐ D. deliberately set below the equilibrium wage to reduce unemployment.

8. What is the relationship between the unemployment rate and enacting or increasing a minimum wage?

When a minimum wage is enacted or increased across a country, the unemployment rate

- ☐ A. will decrease because workers will have more money which will increase demand and cause firms to hire more workers.
- ☐ B. is not affected because the minimum wage does not affect firm hiring decisions.
- ☒ C. will increase some but the impacts will be much larger for some groups of workers.
- ☐ D. will increase and the impacts will be spread evenly across all groups of workers.

9. At the start of the year, the rate of unemployment among a country's citizens who were between 50 to 60 years old was 7 percent. At the end of the year, the same age-group reported an unemployment rate of 6 percent. However, a newspaper columnist claims that the falling unemployment rate is not an accurate indicator of the overall employment situation of this age-group.

Which of the following, if true, would strengthen the columnist's claim?

- ☒ A. Most people in this group who lost their jobs stopped actively looking for new work.
- ☐ B. The overall unemployment in the economy decreased by 2 percent.
- ☐ C. Economic indicators show that the economy is now recovering after a prolonged recession.
- ☐ D. The number of people in this age-group receiving unemployment benefits from the government decreased.
- ☐ E. Most people in this age-group who lost their jobs were able to find other work.

10. Complete the following table:

Product	Base Year			2014		2016	
	Quantity	Price	Expenditures	Price	Expenditures	Price	Expenditures
Denim Jeans	1	\$60	\$60	\$120	\$120	\$95	\$95
Pizzas	20	12	240	18	360	16	320
Coffee Drinks	20	5	100	5	100	6	120
Total		\$	400	\$	580	\$	535

The CPI in 2014 equals . (Enter your response rounded to two decimal places.)

The CPI in 2016 equals . (Enter your response rounded to two decimal places.)

11. What index is used to measure the average prices paid by a typical family?

An average of the prices of the goods and services purchased by a typical family is the:

- ☐ A. inflation rate index.
- ☐ B. producer price index (PPI).
- ☒ C. consumer price index (CPI).
- ☐ D. aggregate price level index.

Suppose the economy's consumer price index (CPI) in 2008 was 190 and the CPI in 2009 was 197.

The inflation rate over the period from 2008-2009 was equal to %. (Enter your response rounded to one decimal place.)

12. The monthly market basket for consumers consists of pizza, t-shirts, and rent.

The table below shows market basket quantities and prices for the base year (Year 1) and in the following two years.

Product	Base Year (Year 1) Quantity	Price in the Base Year	Price in Year 2	Price in Year 3
Pizza	20	\$3.00	\$3.75	\$3.90
T-Shirts	2	\$15.00	\$13.50	\$15.75
Rent	1	\$450.00	\$495.00	\$540.00

The inflation rate between Year 1 and Year 2 is %. (Round both answers to one decimal place.)

The inflation rate between Year 2 and Year 3 is %.

13. The difference between a nominal variable and a real variable is that

- ☐ A. nominal variables are economic variables that are adjusted for inflation, whereas real variables are valued in today's dollars.
- ☐ B. real variables are divided by the price index and multiplied by 100 to obtain nominal variables.
- ☐ C. real variables are calculated in current-year prices and the nominal variables are measured in dollars of the base year to adjust for the effects of inflation.
- ☒ D. nominal variables are calculated in current-year prices and the real variables are measured in dollars of the base year for the price index to correct the effects of inflation.

14. Your father earned \$34,000 per year in 1984. To the nearest dollar, what is that equivalent to in 2014 if the CPI in 2014 is 215 and the CPI in 1984 is 104?

- ☒ A. \$70,288
- ☐ B. \$34,000
- ☐ C. \$73,100
- ☐ D. \$16,447

15. Congratulations! Your boss has given you a raise.

However, you want to know whether your purchasing power has actually increased, since inflation is rising as well. The table below gives you data for wages and the Consumer Price Index (CPI) for the last two years.

	Wage (\$/day)	CPI
Year 1	\$2,500	115
Year 2	\$2,875	124.2

The nominal percentage increase in your wage is %. (Round your answer to one decimal place.)

The real percentage increase in your wage is %. (Round your answer to one decimal place.)

16. In 1972, the price of a Big Mac was \$0.50. Use the information in the table below to find the price of a Big Mac in 2006 dollars.

Year	CPI
1972	41.8
2006	201.6

Source: Bureau of Labor Statistics. Base year = 1962-84.

The value of a 1972 Big Mac in 2006 dollars is \$. (Round your answer to two decimal places.)

The actual price of a Big Mac in 2006 was approximately \$2.99. Thus the price of Big Macs has risen faster than the cost of living.

17. Suppose you were borrowing money to buy a car. Consider the following situations.

Situation 1: Suppose the interest rate on your car loan is 18.00 percent and the inflation rate is 17.00 percent.

The real interest rate is %. (Round your response to two decimal places.)

Situation 2: Suppose the interest rate on your car loan is 5.00 percent and the inflation rate is 2.00 percent.

The real interest rate is %. (Round your response to two decimal places.)

Situation 1 will be better than **Situation 2** because the real interest rate is lower.

18. Suppose that the only good you purchase is hamburgers and that at the beginning of the year, the price of a hamburger is \$2.00. Suppose you lend \$1,000 for one year at an interest rate of 8.3 percent. At the end of the year, hamburgers cost \$2.16.

Calculate the real interest rate¹ you earned on the loan.

The real interest rate is %. (Round your response to one decimal place.)

1: More Info

Real interest rate is the nominal interest rate minus the inflation rate.

19. The true cost of borrowing and lending is best measured by

- ☐ A. the nominal interest rate.
 - ☒ B. the real interest rate.
 - ☐ C. the inflation rate.
 - ☐ D. the three-month Canadian Treasury bill rate.
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20. Suppose that the inflation rate² turns out to be much higher than most people expected. In that case,

- ☐ A. a lender will gain from the situation while a borrower will lose.
- ☐ B. both borrower and lender will lose in this situation.
- ☐ C. both borrower and lender will gain from the situation.
- ☒ D. a borrower will gain from the situation while a lender will lose.

2: Definition

Inflation rate The percentage increase in the price level from one year to the next.

21. Which of the following is a problem of anticipated inflation?

- ☒ A. Increases in the cost of capital for business investment.
 - ☐ B. Redistribution of income from borrowers to lenders.
 - ☐ C. Redistribution of income from lenders to borrowers.
 - ☐ D. Increases in cyclical unemployment.
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22. Fill in the missing values in the table of data collected in a labour force survey in February 2019 for a particular region. (Round your responses for unemployment and labour force to the nearest whole number. Round your response for employment–population ratio to one decimal place.)

Working Age Population	Value
Employment	19,321,100
Unemployment	1454276
Unemployment rate ³	7.0%
Labour force ⁴	20775376
Labour force participation rate ⁵	66.0%
Employment–population ratio ⁶	61.4 %

3: Definition

The unemployment rate is the percentage of the labour force that is unemployed.

4: Definition

The labour force is the sum of employed and unemployed workers in the economy.

5: Definition

The labour force participation rate is the percentage of the working age population in the labour force.

6: Definition

The employment–population ratio is the measure of the portion of the population engaged in paid work.

23. What advice for finding a job would you give someone who is frictionally unemployed⁷?

- ☐ A. Someone who is frictionally unemployed should wait until the recession is over or switch to a part-time job.
- ☐ B. Someone who is frictionally unemployed needs to find another industry to work in.
- ☐ C. Someone who is frictionally unemployed has to acquire the skills necessary to switch industries.
- ☒ D. Someone who is frictionally unemployed has to find an employer who needs their specific skills in the industry they have already worked in.

What advice would you give someone who is structurally unemployed⁸?

- ☒ A. Someone who is structurally unemployed has to acquire the skills necessary to switch industries.
- ☐ B. Someone who is structurally unemployed should wait until the recession is over or switch to a part-time job.
- ☐ C. Someone who is structurally unemployed has to find an employer who needs their specific skills in the industry they have already worked in.
- ☐ D. Someone who is structurally unemployed has no chance of finding a job and should leave the workforce.

What advice would you give someone who is cyclically unemployed⁹?

- ☐ A. Someone who is cyclically unemployed has to find an employer who needs their specific skills in the industry they have already worked in.
- ☐ B. Someone who is cyclically unemployed has to acquire the skills necessary to switch industries.
- ☒ C. Someone who is cyclically unemployed should wait until the recession is over or switch to a part-time job.
- ☐ D. Someone who is cyclically unemployed has no chance of finding a job and should leave the workforce.

7: Definition

Frictional unemployment is short-term unemployment that arises from the process of matching workers with jobs.

8: Definition

Structural unemployment is unemployment that arises from a persistent mismatch between the skills and attributes of workers and the requirements of jobs.

9: Definition

Cyclical unemployment is unemployment caused by a business cycle recession.

24. Which one of the following choices shows how data on nominal wages for 2004 to 2022 and data on the consumer price index¹⁰ for the same years can be used to calculate the real wage for these years?

- ☐ A. $\text{Nominal Wages} - \text{Rate of Inflation} = \text{Real Wages}$
- ☒ B. $\frac{\text{Nominal Wages}}{\text{Price Index}} \times 100 = \text{Real Wages}$
- ☐ C. $\text{Nominal Wages} + \text{Price Index} = \text{Real Wages}$
- ☐ D. $\text{Nominal Wages} \times \text{Price Index} = \text{Real Wages}$

10: Definition

The consumer price index (CPI) is an average of the prices of the goods and services purchased by a typical household.